

PREPARATION LIST

Question 1

Material	Per Student
Prepared yeast (5%) labelled Y	30ml
Ethanol labelled E	25ml
Glucose (0.2M) labelled G	20ml
Methylene blue (0.025%)	6ml
Hot water 80°C (Point to student: either at the side bench in front or at the back of the lab. Lab staff will inform)	Per lab

Apparatus/Item	Per Student
Boiling tubes	5
Small vials	5
Glass beaker for water bath (500ml)	1
Plastic beaker (100ml)	1
Plastic beaker (500ml)	1
Thermometer	1
Stopwatch	1
Syringe (5 cm ³)	4
Dropper (3ml)	1
Distilled water	1
Boiling tube rack	1
Glass rod	1
Marker	1

Question 2

Material	Per Student
Spinach stem	~2 strips
Sucrose solutions: S1 (0.2), S2 (0.4), S3 (0.6), S4(0.8) mol dm ⁻³	Enough for a Petri dish

Apparatus/Item	Per Student
Plasticine	Enough to secure 4 stems
Graph paper	1
Knife	1
White tile	1
Petri dish (without cover)	4
Stopwatch (From Qs 1)	1
Ruler	1
Marker (From Qs 1)	1
Tissue paper	1 roll
Microscope	1
Stage micrometer	1
K1 sample slide	Instr to student: Shared per bench. Odd Seating index no. will use for the first 1hr 15 min followed by the even seating index no. the next 1 hr 15 min.

Instr to students: Extra Reagent can be found on the teacher's bench. Raise your hands to ask for permission before coming to the front to take the reagent.